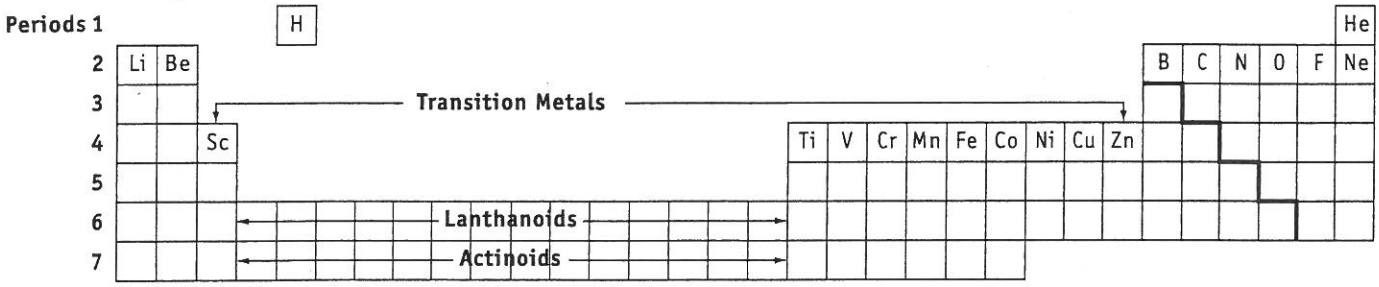
Elements

2 The Periodic Table and organisation of elements

Full Periodic Table

The layout of the full Periodic Table is shown below.



Normal Periodic Table

The Periodic Table is usually in the shorter form shown on page 17.

Periods

- *Periods* are the rows of the Periodic Table. The elements are arranged in order of increasing atomic number (number of protons in each atom's nucleus) across each Period. This is also roughly the order of increasing atomic mass.

1 Complete the following table:

Period	Number of elements in the Period	First element	Last element
1	2		
2			Neon
3			
4			
5		Rubidium	
6			
7	23		Meitnerium

Groups

- *Groups* are the eight main columns. The Groups are usually numbered I–VIII using Roman numerals. The elements in each group have similar properties.
- Group I (alkali metals) begins with the metal lithium. Hydrogen has some chemical properties similar to these metals but because it is a gas it is shown separately.
- Group VII (halogens) comprises very chemically reactive non-metals.
- Group VIII (noble gases) contains gases which are inert (unreactive). These gases are the only elements in which the particles are discrete atoms (separate atoms).

2 True or false?

(a) Sodium and sulfur have similar properties.

(b) Fluorine is chemically similar to chlorine.

(c) A helium balloon contains helium molecules.

(d) Light globes are filled with argon because it does not react with the hot metal filament.

(e) Calcium reacts the same way as magnesium does in many chemical reactions.

- **The zig-zag line in the Periodic Table separates metals from non-metals.** Metals are on the left of the line and non-metals are on the right. Elements close to the line are semi-metals and have both metal and non-metal characteristics.

3 Classify the following elements as metals, semi-metals, or non-metals:

argon	arsenic	boron	bromine	germanium	gold
silicon	silver	sodium	sulfur	tellurium	titanium

- **Transition metals make up the block in the middle of the table.** Many transition metals are used in industry. Compounds of transition metals are often coloured.

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The Periodic Table and organisation of elements (cont.)

4 Name a transition metal which is used:

(a) as the main component of steel _____

(b) in electrical wiring because of its high conductivity _____

(c) in drill bits and light globe filaments because of its hardness and high melting point _____

(d) to make batteries, corrosion-resistant roofing and alloys, e.g. brass _____

(e) where light weight and strength are important, e.g.in spacecraft and artificial joints _____

5 Physical properties Group I

Atomic no. (Z)	Element	Density (g cm^{-3})	Melting point (°C)	Boiling point (°C)
3	Li	0.53	180	1342
11	Na	0.97	98	883
19	K	0.86	63	760
37	Rb	1.53	39	686
55	Cs	1.88	28	669
87	Fr	?	?	?

Francium is a rare, radioactive element isolated in 1939 by Marguerite Perey, who named it after her native country, France. The density, melting point (MP) and boiling point (BP) of francium can be predicted from the trend in these values for the other elements in Group I.

(a) Using graph paper, plot MP against atomic number for the Group I elements. Draw a smooth curve through the points. Extrapolate the curve to find a value for MP at atomic number 87.

(i) Estimated MP of francium = _____ °C

(ii) Write a sentence describing the trend in melting points shown by your graph.

_____

(b) On another sheet of graph paper, plot density value against atomic number for the same elements. Draw a smooth curve which best fits the points. Extrapolate the curve to estimate a value for the density of francium. (Include the unit of density in your answer.)

Estimated density of francium = _____

6 Physical properties Group IV

Atomic no.	Element	Density (g cm^{-3})	Melting point (°C)
6	Carbon		—
	■ diamond allotrope	2.25	(All allotropes sublime;
	■ graphite allotrope	3.51	diamond and graphite
	■ fullerene allotropes	approx. 1.50	sublime at about 4000°C)
14	Silicon	2.33	1420
32	Germanium	5.35	945
50	Tin		
	■ white allotrope	7.28	232
	■ grey allotrope	5.79	—
			(Grey changes to white above 13°C)
82	Lead	11.34	327

(a) (i) Complete the following sentence: _____ are different forms of the same element resulting from different arrangements of the element's atoms.

(ii) In the data given above, what evidence is there that carbon atoms are arranged differently in diamond and graphite? _____

_____

(b) Is there a definite trend in MP in Group IV? Explain. _____

_____

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